The `ltmermaid` package

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Abstract

The `ltmermaid` package renders Mermaid diagrams at LaTeX 2$_\varepsilon$ compile time when using **LuaLaTeX** and embeds them in the PDF. Rendering uses external tools such as the Mermaid CLI, so **shell escape** (`-shell-escape`) is required. The machine that runs `lualatex` must have a working **Node.js** installation, `mmdc` (or `npx`), and headless Chromium for the CLI.

1 Requirements

Item	Requirement
TeX engine	LuaLaTeX (`lualatex`).
Format	LaTeX 2$_\varepsilon$.
Shell escape	`lualatex -shell-escape` (the package invokes the CLI via `os.execute`).
External tool	Mermaid CLI (`mmdc`).

If the Mermaid CLI is missing from the environment, `ltmermaid` does **not** auto-install `mmdc` or similar. A typical setup uses **Node.js** with `npm` (or `npx`). If you pass a command that includes `npx -y` via the package option `Renderer`—for example `npx -y @mermaid-js/mermaid-cli …`—then `npx` may fetch and install packages as needed (**network access** may be required).

When you run `lualatex`, unless the `Renderer` option points to another command, the default renderer `mmdc` must be visible in the same environment (if you set `Renderer`, that command must be visible instead). The Mermaid CLI depends on headless Chromium (Puppeteer); see the Mermaid CLI documentation for details.

2 License and source

Distributed under the LaTeX Project Public License (LPPL), version 1.3c or later.

Source and issues: `https://github.com/ryoya9826/ltMermaid`

3 Usage

3.1 Minimal document

```
\documentclass{article}
\usepackage{ltmermaid}

\begin{document}
\begin{mermaid}
flowchart LR
  A --> B
\end{mermaid}
\end{document}
```

Compile with:

```
lualatex -shell-escape yourfile.tex
```

3.2 Layout adjustments (optional)

By default, included graphics are scaled with `adjustbox`. Example:

```
\MermaidAdjustBoxOpts{max width=0.8\linewidth,center}
\MermaidAdjustBoxOpts{max width=0.9\linewidth,center,valign=T}
```

3.3 Beamer

The package works with the `beamer` class. Any frame that contains a `mermaid` environment must use the `fragile` option (the environment relies on `fancyvrb`), for example `\begin{frame}[fragile]{Diagram}`.

```
\documentclass{beamer}
\usepackage{ltmermaid}

\begin{document}
\begin{frame}[fragile]{Mermaid}
\begin{mermaid}
flowchart LR
  A --> B
\end{mermaid}
\end{frame}
\end{document}
```

4 Package options (optional)

- **`Renderer`**: Prefix for the renderer command. If omitted, the default is `mmdc`. You can set it explicitly, for example `npx -y @mermaid-js/mermaid-cli`.

5 User macros

- **\MermaidRendererOptions{...}:** Extra CLI arguments before -i / -o (merged after -f when it is passed).
- **\MermaidPdfFit / \MermaidNoPdfFit:** Always / never pass -f (--pdfFit) to the renderer. By default, -f is passed only when <renderer> --version reports Mermaid CLI 11 or older (Mermaid CLI 12 removed -f). If the version cannot be detected, -f is not passed; add \MermaidPdfFit if you use Mermaid CLI 11 or older.
- **\MermaidAdjustBoxOpts{...}:** Full adjustbox key list around \includegraphics (default max width=0.9\linewidth,center).
- **\MermaidGraphicsOpts{...}:** Extra keys for \includegraphics (e.g. rotation, trim); width is usually set with \MermaidAdjustBoxOpts.
- **\mermaidfile[<includegraphics options>]{filename}:** Render a Mermaid diagram from an external file (.mmd) instead of inline code. The diagram is rendered and included just like the mermaid environment, and an optional argument may be used to pass keys directly to \includegraphics.

5.1 Including diagrams from files

You can use \mermaidfile[<includegraphics options>]{filename} to render a Mermaid diagram from an external file. Example:

```
1 \mermaidfile[width=0.8\linewidth]{diagram-001.mmd}
```

The optional argument is forwarded directly to \includegraphics, so valid keys include width, height, angle, trim, keepaspectratio, and similar graphics options.

6 Output files

Intermediate .mmd and .pdf files are written under the mermaid/ directory relative to the directory where compilation runs. With -output-directory, they are written under mermaid/ inside the directory indicated by the environment variable TEXMF_OUTPUT_DIRECTORY.

A diagram is rendered only when its source or the renderer settings (Renderer, \MermaidRendererOptions, \MermaidPdfFit / \MermaidNoPdfFit) have changed since the last rendering. To render all diagrams again, delete the mermaid/ directory.

7 Examples

Left: source for the mermaid environment as typed. **Right:** rendered diagram.

```
\begin{mermaid}
flowchart TB
  subgraph client["Client tier"]
    WEB["Browser / SPA"]
    CLI["CLI / batch"]
  end
  subgraph edge["Edge"]
    GW{{API Gateway}}
  end
  subgraph svc["Services"]
    AUTH["Auth"]
    API["Business API"]
    WORK["Workers"]
  end
  subgraph store["Data"]
    DB[("PostgreSQL")]
    CACHE[("Redis")]
    QUEUE["Job queue"]
  end
  WEB --> GW
  CLI --> GW
  GW --> AUTH
  GW --> API
  API --> WORK
  API --> DB
  API --> CACHE
  WORK --> QUEUE
  WORK --> DB
\end{mermaid}
```

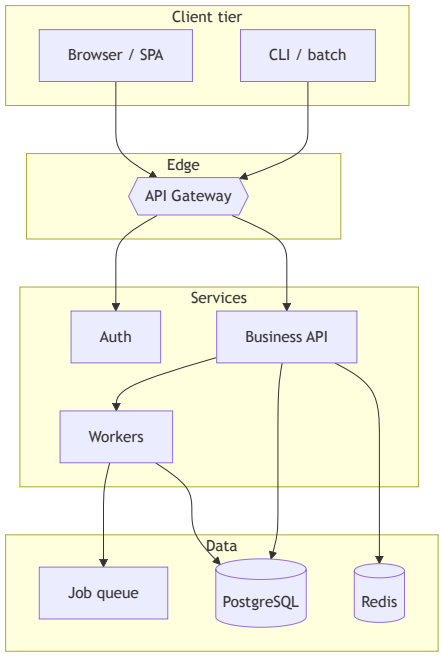


Figure 1: Layered architecture (subgraphs and node shapes)

```
\begin{mermaid}
sequenceDiagram
  autonumber
  actor U as User
  participant B as Browser
  participant A as Auth API
  participant S as Business API
  participant D as DB
  U->>B: Log in
  B->>+A: POST /token
  A->>D: Verify user
  D-->>A: Row
  A-->>-B: JWT
  B->>+S: GET /orders (Bearer)
  S->>A: Validate token
  A-->>S: Claims
  S->>D: SELECT
  D-->>S: Rows
  S-->>-B: 200 JSON
  B-->>U: Show list
\end{mermaid}
```

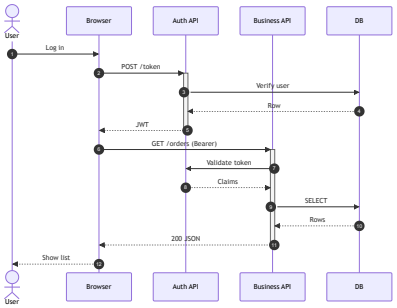


Figure 2: Sequence diagram (autonumber, async arrows, multiple participants)

```
\begin{mermaid}
stateDiagram-v2
  [*] --> Draft: Create
  Draft --> Review: Submit
  Review --> Draft: Send back
  Review --> Approved: Approve
  Approved --> Published: Publish
  Published --> Archived: Close
  Review --> Rejected: Reject
  Rejected --> [*]
  Archived --> [*]
\end{mermaid}
```

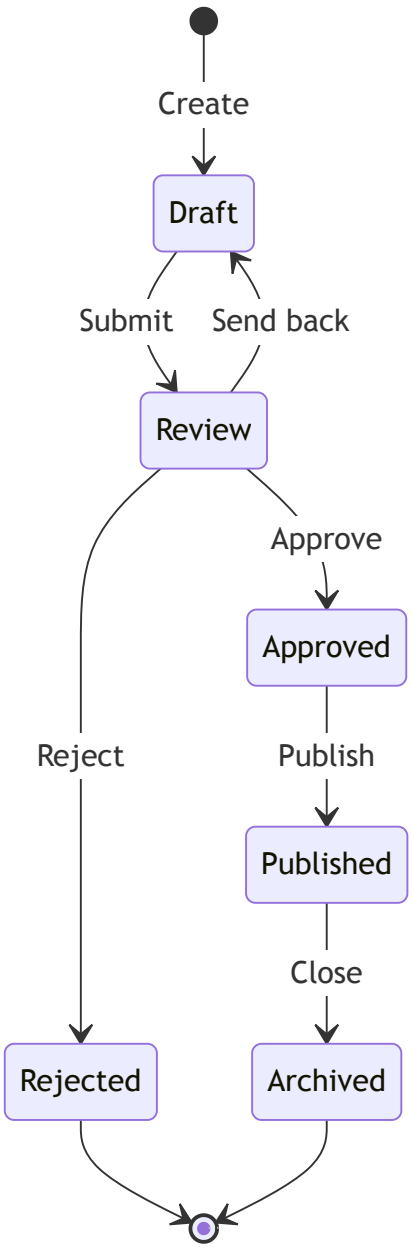


Figure 3: State machine (`stateDiagram-v2`)

8 Version history

Version 1.1 (2026-10-22) Supported Mermaid CLI 12 (`-f` is passed only for Mermaid CLI 11 or older; added `\MermaidPdfFit`), rendered diagrams only when their source or renderer settings change, and fixed

output file name conflicts between the `mermaid` environment and `\mermaidfile`.

Version 1.0c (2026-05-06) Various bug fixes.

Version 1.0b (2026-04-29) Added `\mermaidfile` support, Makefile build/test targets, and `l3build` regression testing.

Version 1.0a (2026-04-20) Renamed sample documents.

Version 1.0 (2026-04-16) Stable release.

Version 0.2 (2026-04-13) Removed `MERMAID_MMDC` and `MERMAID_MMDC_OPTIONS`.

Version 0.1 (2026-04-08) Initial release.